

KONGU ENGINEERING COLLEGE
(AUTONOMOUS)

PERUNDURAI, ERODE – 638 060

EXPERIMENT - 4

PROJECT TITLE

**SMART ELEVATOR AUTOMATION AND HUMAN DETECTOR
SYSTEM USING PIR SENSOR**

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SMART ELEVATOR AUTOMATION AND HUMAN DETECTOR SYSTEM USING PIR SENSOR

1. Project review :

The Smart Elevator Automation and Human Detector System Using PIR Sensor is a compact prototype designed to detect human movement and automatically control the operation of a model elevator system. The system uses an ESP8266 NodeMCU as the central controller and a PIR sensor to detect the presence or movement of a person.

The PIR sensor continuously monitors the surrounding area for human movement. When a person is detected, the PIR sensor sends a signal to the ESP8266 NodeMCU. The controller processes this signal and automatically activates the elevator system. A servo motor can be used to simulate the automatic opening and closing of the elevator door, while a buzzer and LED provide audio and visual indications.

The proposed system provides a simple, low-cost, and automatic method for human detection and elevator automation. It can also be further developed with floor selection, IoT monitoring, mobile notifications, RFID access, and smart building automation.

2. Problem Statement :

Air Conventional elevator systems generally require users to manually press buttons to call or operate the elevator. In some situations, automatic human detection can improve convenience and support smart building automation.

Manual operation may also be inconvenient for elderly persons, persons with disabilities, or people carrying objects. Therefore, there is a need for a simple automated system that can detect the presence of a person and initiate a suitable elevator operation.

The proposed system addresses these problems by integrating a PIR sensor, ESP8266 NodeMCU, servo motor, buzzer, and LED. The PIR sensor detects human movement, while the ESP8266 processes the sensor signal. When a person is detected, the system can automatically open the model elevator door and provide suitable audio and visual indications.

3. Proposed Solution :

The proposed solution consists of an ESP8266-based Air Pollution Monitoring System. The MQ2 gas sensor is used to detect smoke and combustible gases in the surrounding environment and sends the sensor readings to the ESP8266 NodeMCU.

The proposed solution consists of an ESP8266-based Smart Elevator Automation and Human Detector System. The PIR sensor is used to detect human movement near the elevator area and sends a detection signal to the ESP8266 NodeMCU.

The ESP8266 processes the PIR sensor signal and controls the connected components according to the programmed condition. When human movement is detected, the servo motor automatically operates the model elevator door. The LED provides a visual indication, and the buzzer can provide an audible alert.

Proposed Operation:

1. Start the system and initialize the ESP8266.
2. Initialize the PIR sensor, servo motor, buzzer, and LED.
3. The PIR sensor continuously monitors the area.
4. The PIR sensor detects human movement.
5. The PIR sensor sends a detection signal to the ESP8266.
6. ESP8266 processes the sensor signal.
7. If a human is detected, the LED is activated.
8. The buzzer provides an indication.
9. The servo motor opens the model elevator door.
10. After a predefined time delay, the elevator door closes automatically.
11. If no human is detected, the system remains in standby mode.
12. The system continuously repeats the monitoring process.

4. System Architecture :

The Smart Elevator Automation and Human Detector System consists of five main sections:

Input Section:

The input section consists of:

- **PIR Sensor** – Detects human movement in the elevator area.
- The PIR sensor sends the detection signal to the ESP8266 NodeMCU

Processing Section:

The ESP8266 NodeMCU acts as the central controller.

- Receives the human detection signal from the PIR sensor.
- Processes the sensor signal.
- Determines whether a person is detected.
- Controls the servo motor, buzzer, and LED according to the programmed condition.

Actuation Section:

The servo motor is used for automatic model elevator door operation.

- The servo motor receives a control signal from the ESP8266.
- It opens the model elevator door when a person is detected.
- After a predefined delay, it closes the model elevator door automatically.

Indication Section:

The Buzzer and LED provide indications.

- LED OFF and Buzzer OFF → No human detected / standby condition.
- LED ON and Buzzer ON → Human detected / elevator operation activated.

Connectivity Section:

The ESP8266 has built-in Wi-Fi capability.

The system can be further developed for:

- Smart building automation.
- IoT-based elevator monitoring. Mobile notifications.

- Remote system monitoring.
- Floor status monitoring.

5. CIRCUIT TABLE :

Important Pin Mapping:

ESP8266 NodeMCU Pin Component

GPIO 2	D4 PIR Sensor → OUT
GPIO 14	D5 Servo Motor → Signal
GPIO 12	D6 Buzzer → Positive Terminal
GPIO 13	D7 LED → Anode through resistor
5V / VIN	– PIR Sensor and Servo Power
GND	– Common Ground

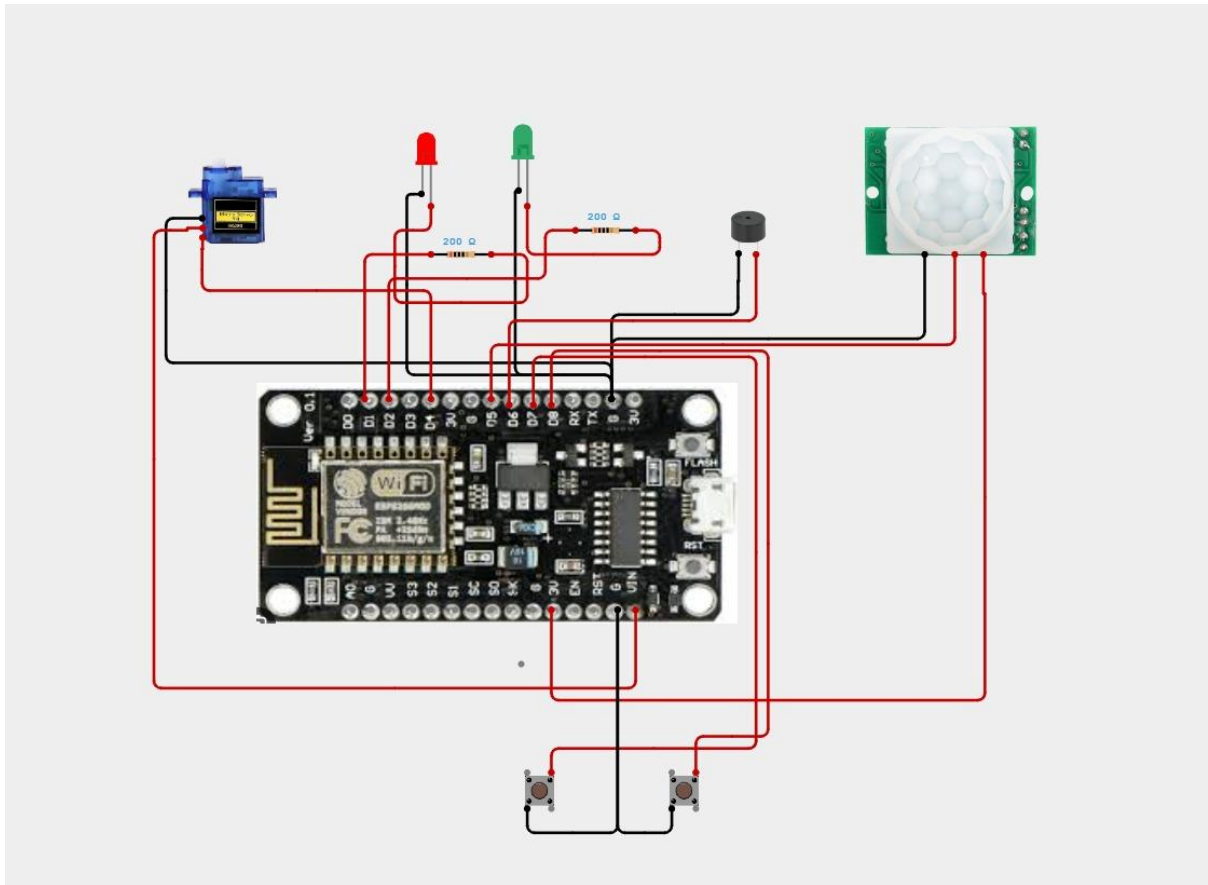
6. Circuit Design :

The circuit consists of an ESP8266 NodeMCU, PIR sensor, servo motor, buzzer, LED, breadboard, and jumper wires.

The PIR sensor is connected to the ESP8266 to detect human movement near the model elevator. The output pin of the PIR sensor is connected to GPIO2 (D4) of the ESP8266.

The servo motor is connected to GPIO14 (D5) and is used to simulate the automatic opening and closing of the elevator door. The buzzer is connected to GPIO12 (D6) and provides an audible indication when a person is detected. The LED is connected to GPIO13 (D7) through a suitable resistor and provides a visual indication.

The ESP8266 acts as the main controller and continuously monitors the PIR sensor. When human movement is detected, the ESP8266 activates the LED and buzzer and operates the servo motor to open the elevator door. After a short delay, the door automatically closes.



8. Coding :

// SMART ELEVATOR AUTOMATION AND HUMAN DETECTION SYSTEM

// ESP8266 NodeMCU + PIR + Servo + LEDs + Buzzer

// + UP/DOWN Push Buttons

// =====

#include <Servo.h>

// =====

```
// PIN DEFINITIONS
```

```
// =====
```

```
// PIR Motion Sensor
```

```
#define PIR_PIN D5
```

```
// SG90 Servo
```

```
#define SERVO_PIN D4
```

```
// Green LED
```

```
#define GREEN_LED D1
```

```
// Red LED
```

```
#define RED_LED D2
```

```
// Buzzer
```

```
#define BUZZER_PIN D6
```

```
// UP Push Button
```

```
#define UP_BUTTON D7
```

```
// DOWN Push Button
```

```
#define DOWN_BUTTON D8
```

```
// =====
```

```
// SERVO OBJECT
```

```
// =====
```

```
Servo elevatorServo;
```

```
// =====
```

```
// SETUP
```

```
// =====
```

```
void setup()
```

```
{
```

```
  Serial.begin(9600);
```

```
  // PIR
```

```
  pinMode(PIR_PIN, INPUT);
```

```
  // LEDs
```

```
  pinMode(GREEN_LED, OUTPUT);
```

```
  pinMode(RED_LED, OUTPUT);
```

```
  // Buzzer
```

```
  pinMode(BUZZER_PIN, OUTPUT);
```

```
  // Push buttons
```

```
  pinMode(UP_BUTTON, INPUT_PULLUP);
```

```
  pinMode(DOWN_BUTTON, INPUT_PULLUP);
```

```
// Servo
elevatorServo.attach(SERVO_PIN);

// Initial elevator position
elevatorServo.write(0);

// Initial status
digitalWrite(GREEN_LED, LOW);
digitalWrite(RED_LED, HIGH);
digitalWrite(BUZZER_PIN, LOW);

Serial.println();
Serial.println("=====");
Serial.println(" SMART ELEVATOR AUTOMATION SYSTEM");
Serial.println("=====");
Serial.println("System Initializing...");

delay(3000);

Serial.println("System Ready");
Serial.println("-----");
}

// =====
// LOOP
// =====
```

```
void loop()
{
    int pirState = digitalRead(PIR_PIN);

    // =====
    // HUMAN DETECTION
    // =====

    if (pirState == HIGH)
    {
        Serial.println("Human Detected!");
        Serial.println("Elevator Door Opening...");

        // Green LED ON
        digitalWrite(GREEN_LED, HIGH);

        // Red LED OFF
        digitalWrite(RED_LED, LOW);

        // Buzzer indication
        digitalWrite(BUZZER_PIN, HIGH);
        delay(300);
        digitalWrite(BUZZER_PIN, LOW);

        // Open elevator door
```

```
elevatorServo.write(90);
```

```
delay(3000);
```

```
Serial.println("Elevator Door Open");
```

```
Serial.println("-----");
```

```
}
```

```
else
```

```
{
```

```
    // No human detected
```

```
    digitalWrite(GREEN_LED, LOW);
```

```
    digitalWrite(RED_LED, HIGH);
```

```
elevatorServo.write(0);
```

```
Serial.println("No Human Detected");
```

```
Serial.println("Elevator Door Closed");
```

```
}
```

```
// =====
```

```
// UP BUTTON
```

```
// =====
```

```
if (digitalRead(UP_BUTTON) == LOW)
```

```
{
```

```
    Serial.println("UP Button Pressed");
```

```
Serial.println("Elevator Moving UP");
```

```
digitalWrite(BUZZER_PIN, HIGH);
```

```
delay(200);
```

```
digitalWrite(BUZZER_PIN, LOW);
```

```
digitalWrite(GREEN_LED, HIGH);
```

```
digitalWrite(RED_LED, LOW);
```

```
elevatorServo.write(90);
```

```
delay(2000);
```

```
Serial.println("UP Operation Completed");
```

```
Serial.println("-----");
```

```
}
```

```
// =====
```

```
// DOWN BUTTON
```

```
// =====
```

```
if (digitalRead(DOWN_BUTTON) == LOW)
```

```
{
```

```
Serial.println("DOWN Button Pressed");
```

```
Serial.println("Elevator Moving DOWN");
```

```
digitalWrite(BUZZER_PIN, HIGH);
```

```
delay(200);
```

```
digitalWrite(BUZZER_PIN, LOW);
```

```
digitalWrite(GREEN_LED, HIGH);
```

```
digitalWrite(RED_LED, LOW);
```

```
elevatorServo.write(90);
```

```
delay(2000);
```

```
Serial.println("DOWN Operation Completed");
```

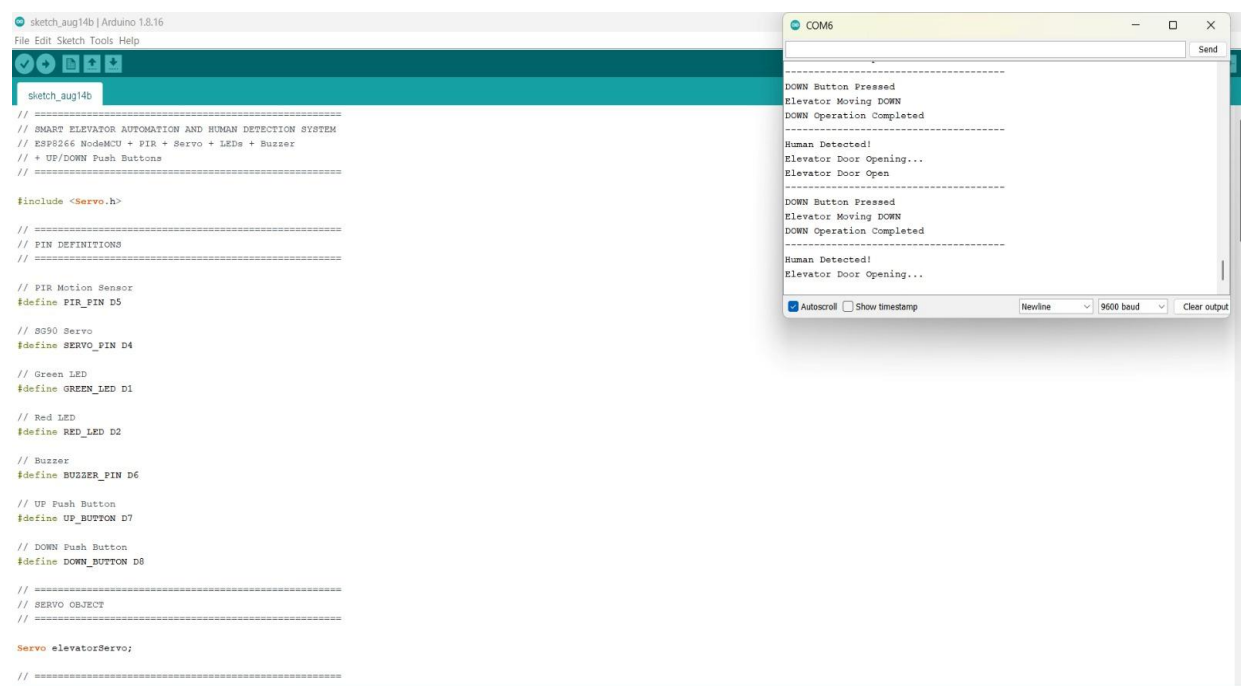
```
Serial.println("-----");
```

```
}
```

```
delay(500);
```

```
}
```

CODING OUTPUT:



```
sketch_aug14b | Arduino 1.8.16
File Edit Sketch Tools Help

sketch_aug14b

// =====
// SMART ELEVATOR AUTOMATION AND HUMAN DETECTION SYSTEM
// ESP8266 NodeMCU + PIR + Servo + LEDs + Buzzer
// + UP/DOWN Push Buttons
// =====

#include <Servo.h>

// =====
// PIN DEFINITIONS
// =====

// PIR Motion Sensor
#define PIR_PIN D5

// SG90 Servo
#define SERVO_PIN D4

// Green LED
#define GREEN_LED D1

// Red LED
#define RED_LED D2

// Buzzer
#define BUZZER_PIN D6

// UP Push Button
#define UP_BUTTON D7

// DOWN Push Button
#define DOWN_BUTTON D8

// =====
// SERVO OBJECT
// =====

Servo elevatorServo;

// =====
```

```
COM6
Send

-----
DOWN Button Pressed
Elevator Moving DOWN
DOWN Operation Completed
-----
Human Detected!
Elevator Door Opening...
Elevator Door Open
-----
DOWN Button Pressed
Elevator Moving DOWN
DOWN Operation Completed
-----
Human Detected!
Elevator Door Opening...
-----
Autoscroll Show timestamp Newline 9600 baud Clear output
```

9. System Working :

Power ON the System

The ESP8266 NodeMCU is powered ON.

Initialize Components

The PIR sensor, servo motor, buzzer, and LED are initialized.

Detect Gas and Smoke

When a person moves within the detection range, the PIR sensor detects the movement

Send Sensor Data

The PIR sensor sends a HIGH signal to the ESP8266 NodeMCU.

Process the Data

The ESP8266 receives and processes the human detection signal.

Human Detected Condition

If human movement is detected:

- LED → ON
- Buzzer → ON for a short duration
- Servo Motor → Opens the elevator door
- Elevator automation → Activated

Normal Standby Condition

If no human movement is detected:

- LED → OFF
- Buzzer → OFF
- Elevator door → Closed
- System → Standby mode

Repeat the Process

- The PIR sensor continuously monitors the area, and the ESP8266 repeats the automation process.

10. PROTOTYPE IMPLEMENTATION :

The prototype is implemented using an ESP8266 NodeMCU, PIR sensor, servo motor, buzzer, LED, breadboard, and jumper wires. The PIR sensor is placed near the model elevator entrance to detect human movement.

Hardware Setup

1. Connect the PIR sensor OUT pin → D4 (GPIO2) of ESP8266.
2. Connect the Servo Motor signal pin → D5 (GPIO14).
3. Connect the Buzzer → D6 (GPIO12).
4. Connect the LED → D7 (GPIO13) through a suitable resistor.
5. Connect the PIR sensor VCC → suitable power supply.
6. Connect the Servo Motor VCC → suitable power supply.
7. Connect all GND terminals to a common ground.
8. Connect all required low-voltage components using jumper wires.

9. Upload the program to the ESP8266 using the Arduino IDE.
10. Power ON the prototype.
11. Move a person or object within the PIR sensor detection range.
12. Observe the automatic activation of the LED, buzzer, and servo motor.
13. After the predefined delay, observe the automatic closing of the model elevator door.

Prototype Operation

The prototype demonstrates automatic human detection and elevator door automation. The PIR sensor detects human movement, while the ESP8266 processes the detection signal and controls the servo motor, buzzer, and LED according to the programmed condition.

Prototype Working :

- The ESP8266 NodeMCU is powered ON and initializes all connected components.
- The PIR sensor continuously monitors the elevator entrance area.
- When human movement is detected, the PIR sensor sends a signal to the ESP8266.
- The ESP8266 processes the sensor signal.
- The LED turns ON to indicate that a person has been detected.
- The buzzer provides an audible indication.
- The servo motor connected to D5 (GPIO14) operates to open the model elevator door.
- The door remains open for a predefined time.
- After the delay, the servo motor automatically closes the elevator door.
- The LED and buzzer return to their normal OFF condition.
- The system continues monitoring for further human movement.

11. Bill of Materials :

S.No.	Component	Specification	Quantity
1	ESP8266	NodeMCU Wi-Fi Development Board	1
2	PIR Sensor	Human Motion Detection Sensor	1
3	Servo Motor	SG90 or Equivalent Micro Servo	1
4	Buzzer	3.3V/5V Buzzer	1
5	LED	Standard Indicator LED	1
6	Resistor	Suitable LED Current Limiting Resistor	1
7	Breadboard	Standard Prototype Board	1
8	Jumper Wires	Male-Male / Male-Female	1 Set
9	USB Cable	Micro-USB	1
10	Power Supply	Suitable Regulated Supply	1

12. Results & Performance Analysis :

- The PIR sensor successfully detects human movement near the elevator area.
- The ESP8266 NodeMCU successfully receives and processes the PIR sensor signal.
- The system provides continuous monitoring for human movement.
- When a person is detected, the LED provides a visual indication.
- The buzzer provides an audible indication.
- The servo motor successfully demonstrates automatic opening and closing of the model elevator door.
- When no human movement is detected, the system remains in standby mode.

- The system provides an automatic and continuous human detection and elevator automation process.

Results :

- The MQ2 gas sensor successfully detects the presence of smoke and combustible gases.
- The ESP8266 NodeMCU successfully receives and processes the sensor data.
- The system continuously monitors the gas level in the surrounding environment.
- When the gas level exceeds the predefined safety limit, the buzzer turns ON.
- The LED provides a visual warning indication during an unsafe air condition.
- The relay activates to demonstrate ventilation or exhaust control.
- When the gas level returns to the normal range, the buzzer and LED turn OFF.
- The prototype demonstrates the basic operation of an Air Pollution Monitoring System.

Performance Analysis

Parameter	Expected Performance
Human Detection Continuous	Human Detection Continuous
PIR Sensor Real-time Response Detection	PIR Sensor Real-time Response Detection
Human Detected LED ON	Human Detected LED ON
Audio Indication Buzzer Activated	Audio Indication Buzzer Activated
Door Operation Servo Motor Activated	Door Operation Servo Motor Activated
Door Closing Automatic After Delay	Door Closing Automatic After Delay